



Department of Applied Sciences

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What is DevOps?

DevOps is a cultural and technical movement that combines software development (Dev) and IT operations (Ops) to shorten the systems development life cycle while delivering features, fixes and updates frequently in close alignment with business objectives. It emphasises collaboration, automation, continuous integration, continuous delivery, and monitoring throughout the software development process.

Core Principles of DevOps:

- Collaboration: Breaking down silos between development and operations teams
- Automation: Automating repetitive tasks to reduce errors and increase efficiency
- Continuous Integration/Continuous Deployment (CI/CD): Frequent, reliable releases
- Monitoring and Feedback: Continuous monitoring and rapid feedback loops
- Infrastructure as Code: Managing infrastructure through code and version control

THE SEVEN PHASES OF DEVOPS

1. PLAN

Purpose: Define project requirements, goals, and success metrics

Activities:

- Requirements gathering and analysis
- Project planning and resource allocation
- Risk assessment and mitigation strategies
- Setting up project management tools and workflows

Tools: Jira, Confluence, Azure DevOps, GitHub Projects

2. CODE

Purpose: Develop software following best practices and coding standards

Activities:

- Writing clean, maintainable code
- Code reviews and pair programming
- Version control management
- Following coding standards and conventions

Tools: Git, GitHub, GitLab, Bitbucket, IDEs (VS Code, IntelliJ)

3. BUILD

Purpose: Compile, package, and prepare software for deployment

Activities:

- Source code compilation
- Dependency management
- Artifact creation and packaging
- Build optimization and caching

Tools: Maven, Gradle, npm, Docker, Jenkins, Azure Pipelines

4. TEST

Purpose: Ensure software quality through comprehensive testing

Activities:

- Unit testing
- Integration testing
- Performance testing
- Security testing
- Automated test execution

Tools: JUnit, Selenium, Jest, Postman, SonarQube

5. RELEASE

Purpose: Deploy software to production environments safely and reliably

Activities:

- Release planning and coordination
- Environment preparation
- Deployment automation
- Rollback procedures

Tools: Jenkins, GitLab CI/CD, Azure DevOps, Kubernetes

6. DEPLOY

Purpose: Make software available to end users

Activities:

- Production deployment
- Configuration management
- Service orchestration
- Blue-green or canary deployments

Tools: Docker, Ansible, Terraform, AWS/Azure/GCP

7. OPERATE

Purpose: Monitor, maintain, and optimize software in production

Activities:

- System monitoring and alerting
- Performance optimization
- Incident response and troubleshooting
- Capacity planning

Tools: Prometheus, Grafana, ELK Stack, New Relic, Datadog

IS DEVOPS A SOFTWARE PROCESS MODEL?

DevOps can be considered a software process model, but with important distinctions:

Characteristics that make it a process model:

- Structured phases: Clear sequence of activities
(Plan - Code - Build - Test - Release - Deploy - Operate)
- Repeatable process: Consistent methodology across projects
- Defined activities: Specific tasks and deliverables for each phase
- Tool integration: Standardized tools and practices
- Quality gates: Checkpoints and criteria for progression

How it differs from traditional process models:

- Continuous nature: Unlike waterfall, phases overlap and iterate
- Cultural emphasis: Focus on collaboration and mindset, not just process
- Automation-driven: Heavy reliance on automation tools
- Feedback loops: Continuous monitoring and improvement
- Cross-functional: Breaks traditional organizational boundaries

DEVOPS VS. AGILE: COMPARISON AND CONTRAST

SIMILARITIES:

- Iterative approach: Both use continuous cycles (DevOps: continuous delivery, Agile: sprints)
- Collaboration: Both emphasise cross-functional team collaboration
- Customer focus: Both prioritise rapid feature delivery and customer feedback
- Flexibility: Both are adaptable to changes and responsive to requirements
- Quality emphasis: Both integrate continuous testing and quality assurance

KEY DIFFERENCES:

Scope and Focus:

- DevOps: Focuses on the entire software delivery pipeline (development to operations)
- Agile: Primarily focuses on software development methodology

Timeline:

- DevOps: Continuous, 24/7 operations mindset
- Agile: Time-boxed sprints (typically 1-4 weeks)

Primary Concerns:

- DevOps: Deployment, monitoring, infrastructure, reliability
- Agile: Requirements, development, testing, customer satisfaction

Team Structure:

- DevOps: Breaks down Dev-Ops silos, includes operations teams
- Agile: Development teams with product owners and scrum masters

Deliverables:

- DevOps: Working software in production
- Agile: Potentially shippable product increments

Tools and Practices:

- DevOps: CI/CD pipelines, infrastructure automation, monitoring tools
- Agile: Scrum boards, user stories, sprint planning tools

INTEGRATION:

DevOps and Agile are complementary, not competing methodologies:

- Agile provides the development framework
- DevOps extends Agile principles to operations and deployment
- Together, they create a comprehensive approach to software delivery

CONCLUSION:

DevOps represents an evolution in software development that addresses the gap between development and operations. While it functions as a software process model with its seven distinct phases, it's more accurately described as a cultural and technical movement that emphasises collaboration, automation, and continuous improvement. When combined with Agile methodologies, DevOps creates a powerful framework for delivering high-quality software rapidly and reliably.

The seven phases of DevOps provide a structured approach to software delivery while maintaining the flexibility and responsiveness that modern software development demands. This makes DevOps not just a process model, but comprehensive philosophy that transforms how organizations approach software development and operations.

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